

1. Objective of Turtle games

The objective of turtle games is to improve overall sales performance by utilising customer trends.

2. Objective of this report

The objective of this report is to analyse the provided dataset to find the evident customer trends in the dataset to provide actionable recommendations to turtle games to improve overall sales performance.

3. Problem question:

How can Turtle games use the customer trends to improve overall sales performance?

4. Analytical approach

For the purpose of analysis, in this report, age, remuneration and spending score columns have been used as an independent variable to find correlations with loyalty points column. This has been done in order to find relationship between the independent variables and the accumulation of loyalty points by the customer.

Summary and review columns have been used to find the customer sentiment towards the offerings of turtle games.

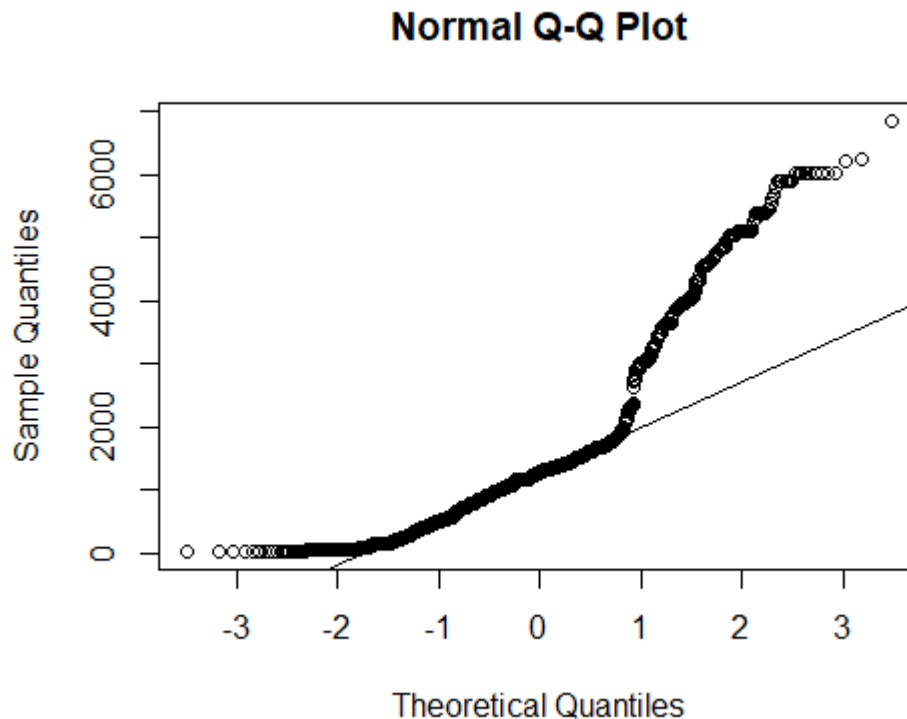
5. Descriptive statistics

gender	age	remuneration	spending_score	loyalty_points	education	product
Length:2000	Min. :17.00	Min. : 12.30	Min. : 1	Min. : 25	Length:2000	Min. : 107
Class :character	1st Qu.:29.00	1st Qu.: 30.34	1st Qu.:32	1st Qu.: 772	Class :character	1st Qu.: 1589
Mode :character	Median :38.00	Median : 47.15	Median :50	Median :1276	Mode :character	Median : 3624
	Mean :39.49	Mean : 48.08	Mean :50	Mean :1578		Mean : 4321
	3rd Qu.:49.00	3rd Qu.: 63.96	3rd Qu.:73	3rd Qu.:1751		3rd Qu.: 6654
	Max. :72.00	Max. :112.34	Max. :99	Max. :6847		Max. :11086
review	summary	remuneration_range	spending_score_quantile			
Length:2000	Length:2000	Low :677	Low :515			
Class :character	Class :character	Medium:679	Medium-Low :510			
Mode :character	Mode :character	High :644	Medium-High:525			
			High :450			

Minimum age of the customer is 17 years and maximum age of the customer is 72 years, average age of the customer is 39 years. Minimum remuneration of the customer per year is £12,300, maximum remuneration is £112,340, average remuneration per customer is £48,080. Minimum spending score of customer is 1, maximum spending score is 99,

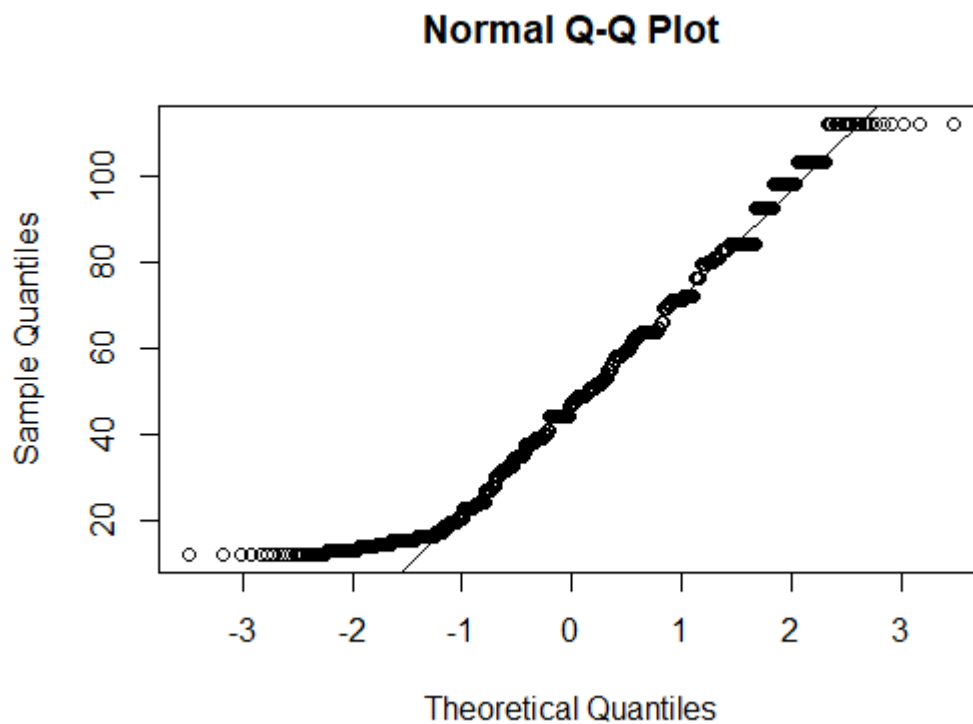
average spending score is 50. Minimum loyalty points accumulated by customer is 25, maximum loyalty points accumulated by customer is 6847, average loyalty points per customer is 1578.

5.1 Shape of loyalty points data against normal distribution



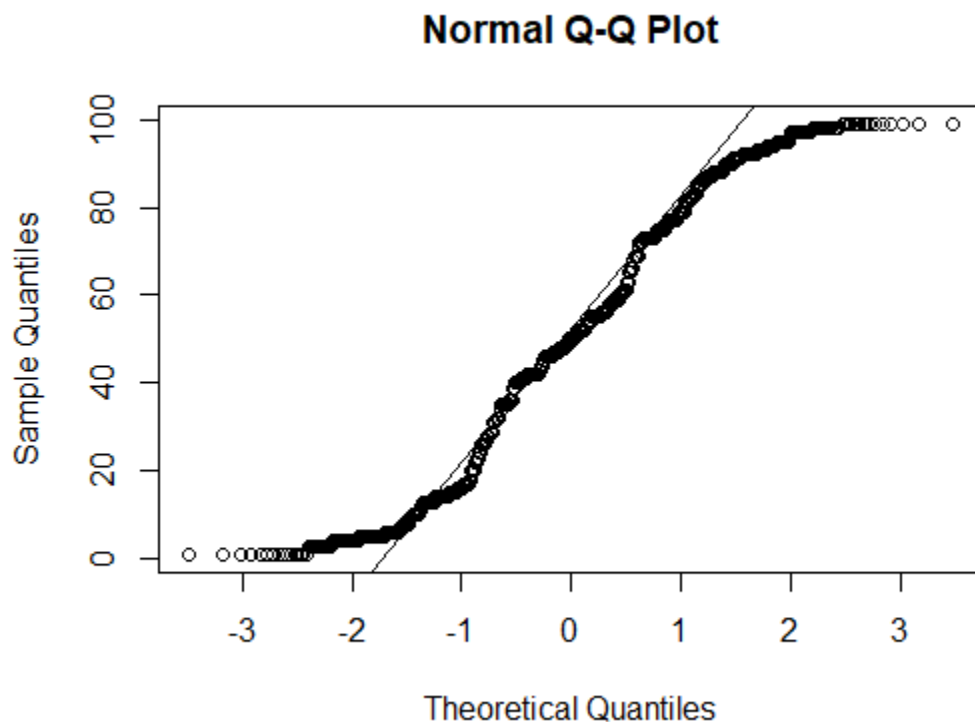
The findings show the loyalty points data is not normally distributed. The skewness is a positive score of 1.46 meaning the tail is more pronounced on the right. This suggests that most values in the loyalty points data are found on the left side of the mean of 1578. The most extreme values are found further to the left. Kurtosis of 4.70 suggest that the loyalty points data is heavy-tailed meaning that most of the contribution to the variance is coming from a handful of extreme data points.

5.2 Shape of remuneration data against normal distribution



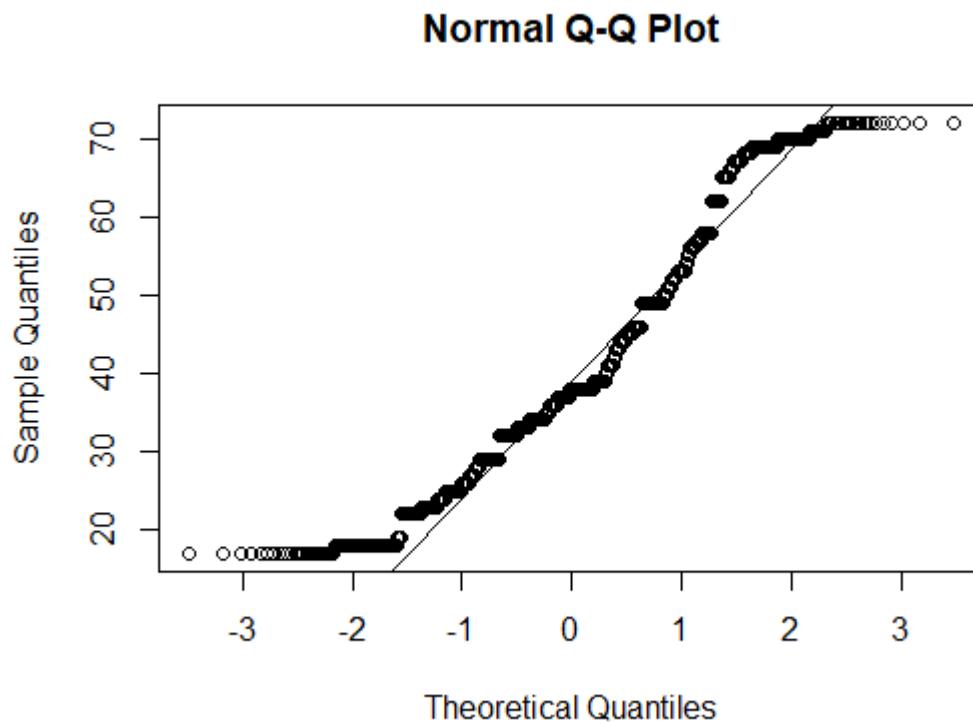
The findings show the remuneration data is not normally distributed. The skewness is a positive score of 0.41 meaning the tail is more pronounced on the right. This suggests that most values in the remuneration data are found on the left side of the mean(48.08). The most extreme values are found further to the left. Kurtosis of 2.59 suggest that the age data is light-tailed.

5.3 Shape of spending score data against normal distribution



The findings show the spending score data is not normally distributed. The skewness is a negative score of -0.041 meaning the tail is more pronounced on the left. This suggests that most values in the spending score data are found on the right side of the mean(50). The most extreme values are found further to the left. Kurtosis of 2.11 suggest that the spending score data is light-tailed.

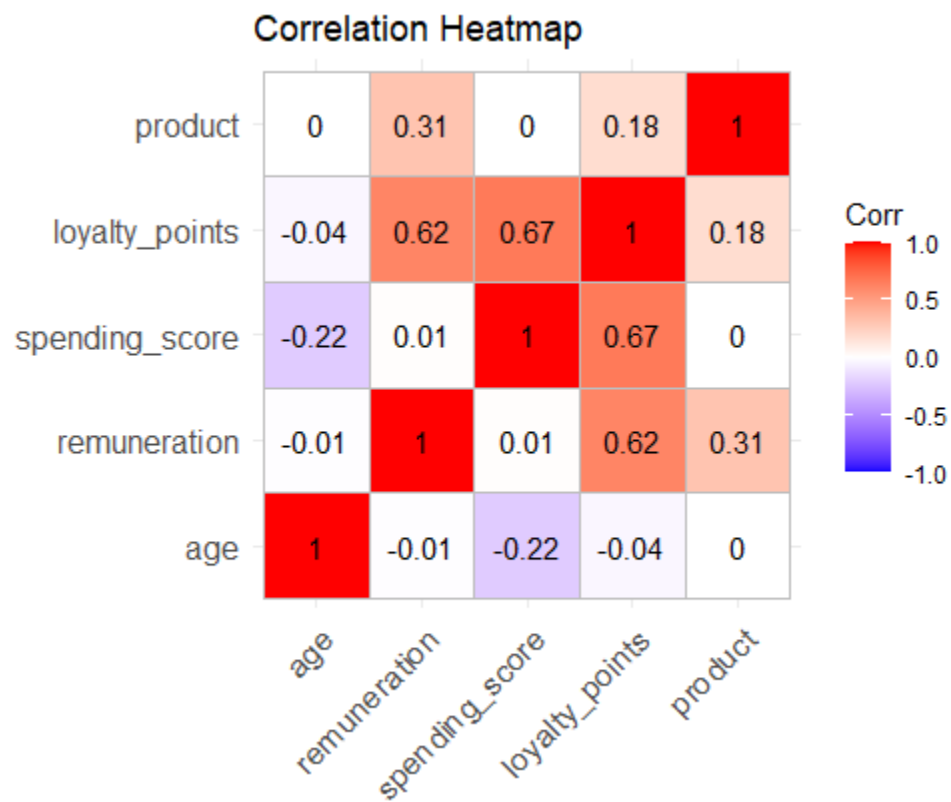
5.4 Shape of age data against normal distribution



The findings show the age data is not normally distributed. The skewness is a positive score of 0.608 meaning the tail is more pronounced on the right. This suggests that most values in the age data are found on the left side of the mean(39.49). The most extreme values are found further to the left. Kurtosis of 2.80 suggest that the age data is light-tailed.

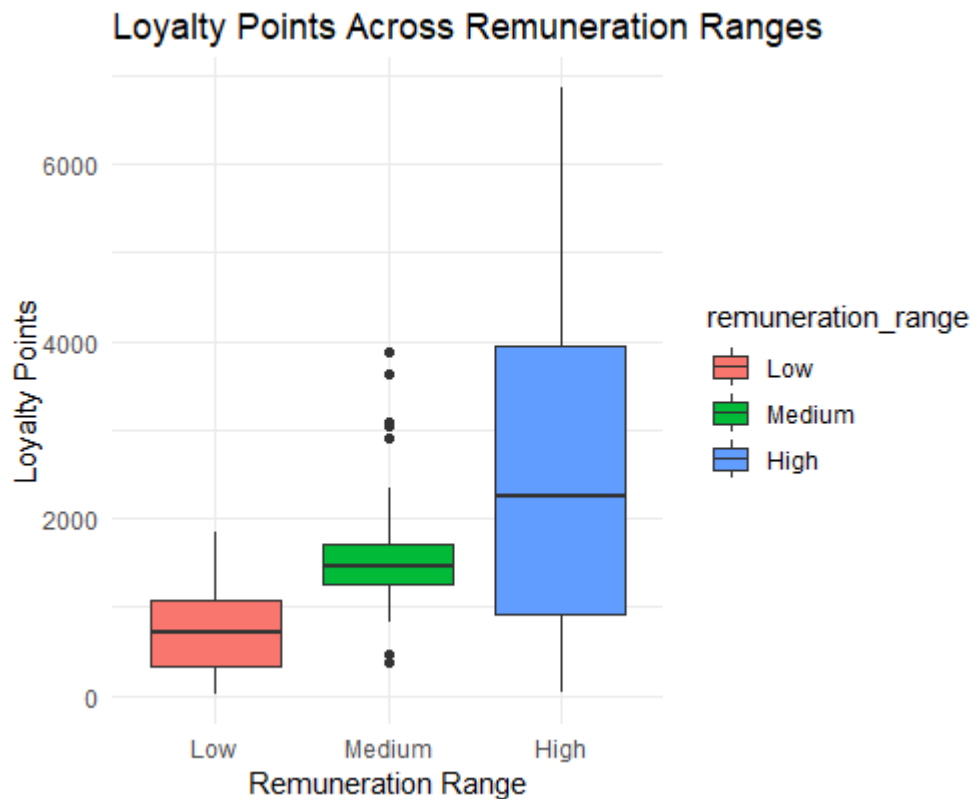
6. Findings

6.1 Correlation Heatmap



Moderate positive correlations are seen between remuneration and loyalty points(.62), and between spending score and loyalty points(.67). Other correlations are either weak or negligible.

6.2 Loyalty points accumulation across remuneration range



The mean loyalty points increase significantly as remuneration increases:

1. Low earners: Mean loyalty points = 725
2. Medium earners: Mean loyalty points = 1551
3. High earners: Mean loyalty points = 2503

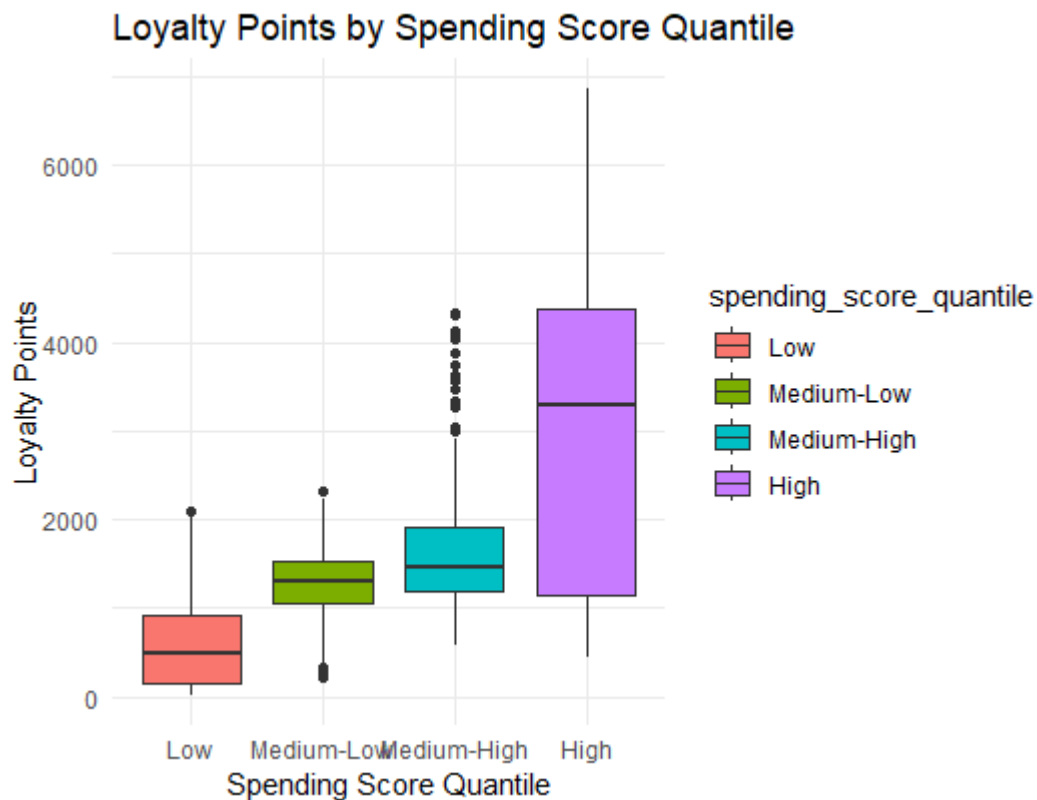
This suggests that higher earners tend to accumulate more loyalty points on average.

The median loyalty points show a similar trend:

1. Low earners: Median loyalty points = 724
2. Medium earners: Median loyalty points = 1463
3. High earners: Median loyalty points = 2262

This reinforces the observation that higher earners are accumulating more loyalty points, and it seems to be consistent across both the mean and median values.

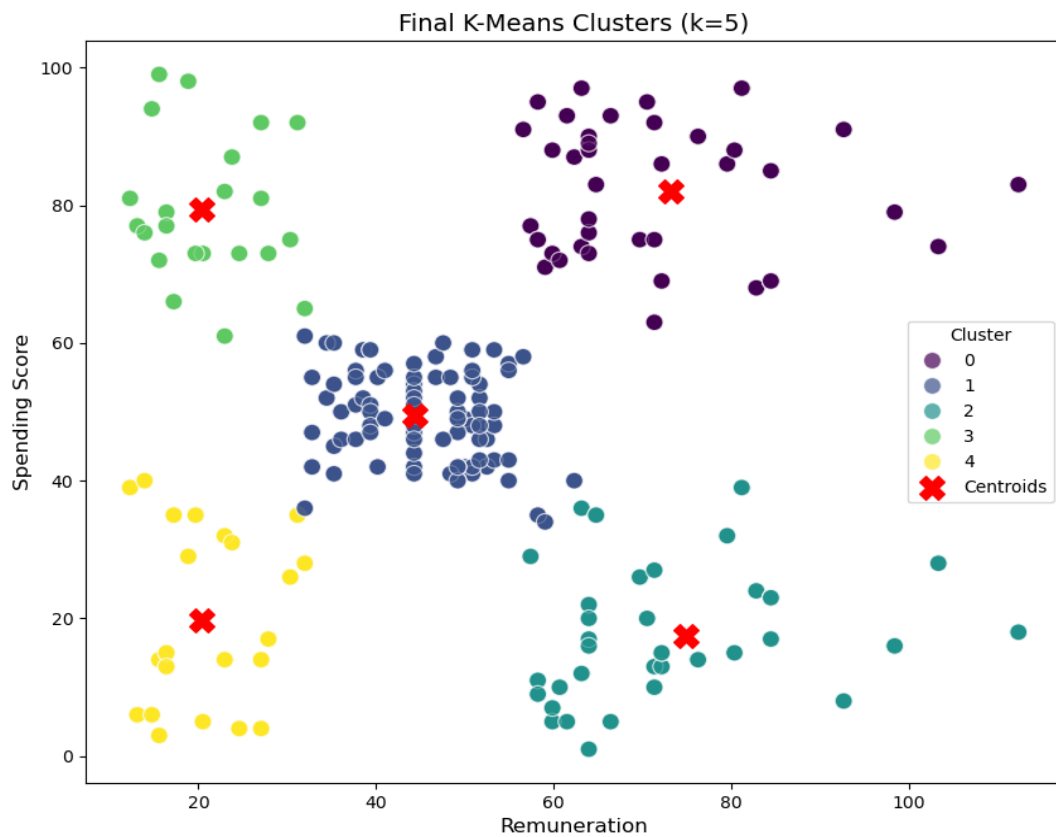
6.3 Loyalty points accumulation by spending score



The data shows that customers in the High spending group tend to accumulate more loyalty points. From the result, it can be seen that there is a clear trend where customers in the High spending group have significantly higher mean and median loyalty points compared to those in the Low spending group. This indicates that the loyalty program at turtle games is rewarding higher spenders with more loyalty points.

Both the Medium-Low and Medium-High groups have a relatively higher range of loyalty points compared to the Low group. Medium-High group, in particular, shows a good balance between mean and median loyalty points, suggesting more consistent loyalty point accumulation. In the Low spending group, there are some customers who accumulate disproportionately high loyalty points.

6.4 Customer Segmentation based on Remuneration and spending



```
# Check the number of observations in each cluster
cluster_counts = df3['cluster'].value_counts()

# Print the count of observations per cluster
print(cluster_counts)
```

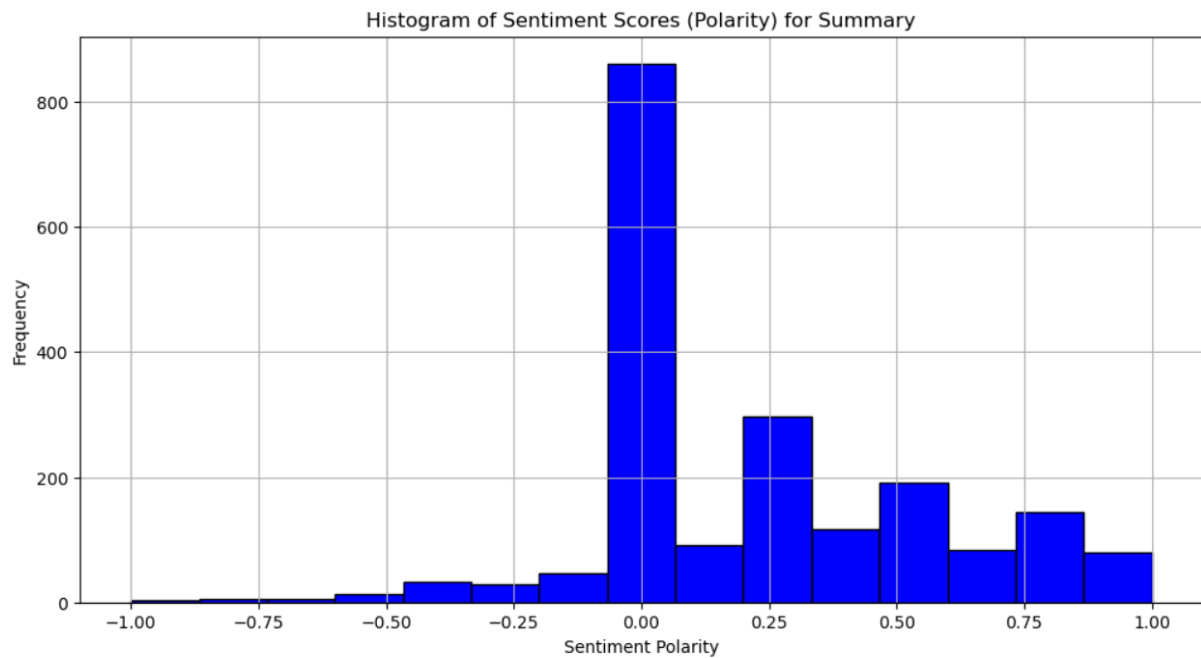
```
cluster
1      774
0      356
2      330
4      271
3      269
Name: count, dtype: int64
```

Based on remuneration and spending patterns, five clusters of customer group have been found to be present in the dataset.

CLUSTER	Characteristics	Customer size
0	High income, high spending: This cluster of customers are heavy spenders and earn high remuneration	Second largest customer group
1	Middle income, moderate spending: This cluster of customers have moderate remuneration and spend moderately.	Largest customer group
2	High income, low spending: This cluster of customers have high remuneration but are low spenders.	Third largest customer group
3	Low income, high spending: This cluster of customers have low remuneration but are high spenders.	Smallest customer group
4	Low income, low spending: This cluster of customers have low remuneration and are also low spenders.	Second smallest customer group.

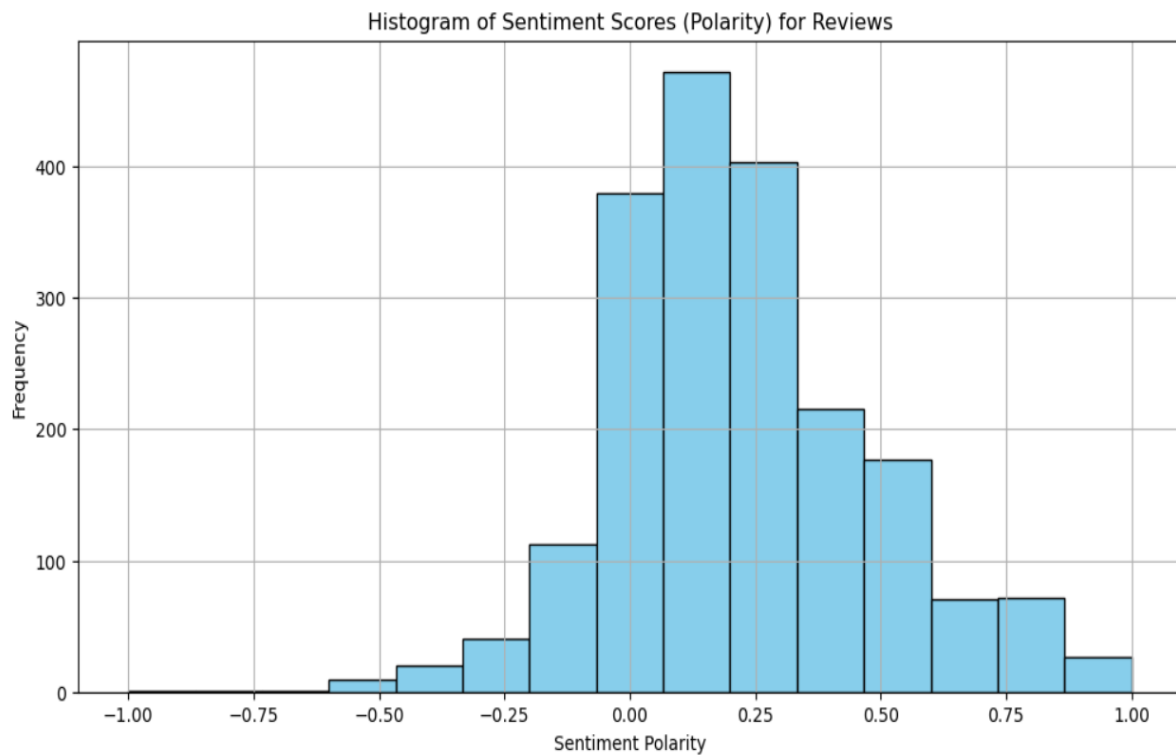
6.5 Sentiment analysis

6.5.1 Histogram of sentiment scores(Polarity) for summary



The result suggest that overall, the summaries are somewhat positive, but not strongly positive. A score of 0.22 indicates that the text conveys a generally favourable tone towards Turtle games.

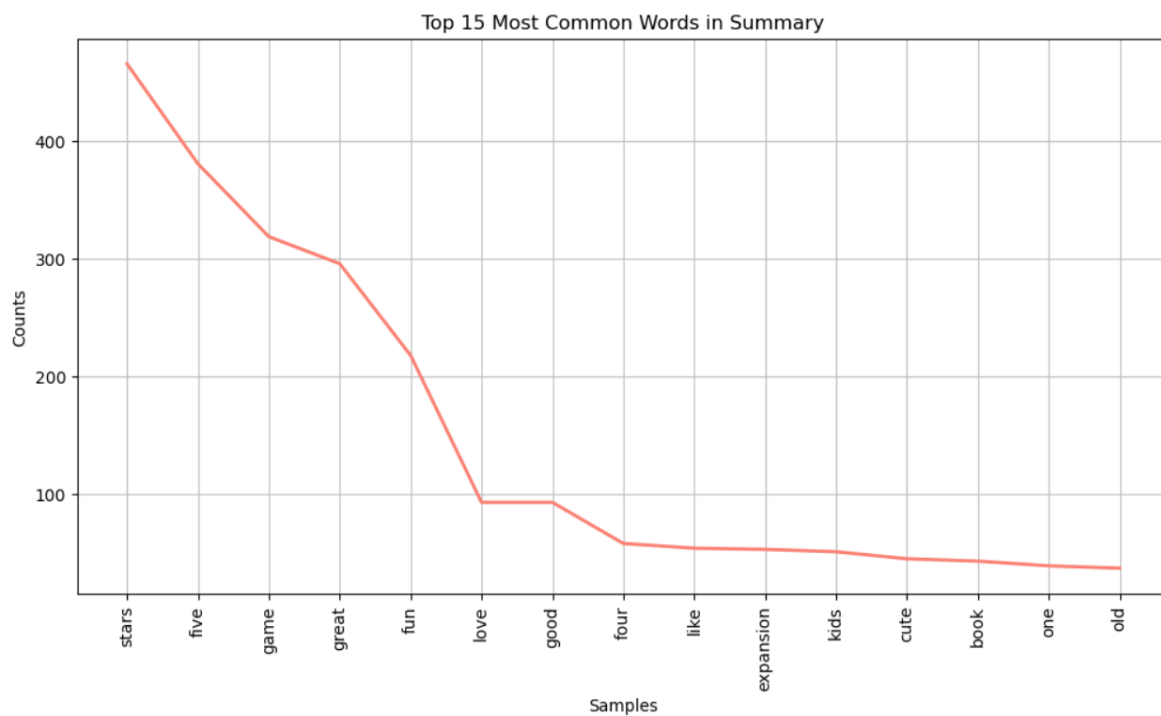
6.5.2 Histogram of sentiment scores(polarity) for reviews



Mean sentiment polarity for reviews: 0.2169495173987102

The mean sentiment polarity for the reviews column is 0.2170, which indicates that, on average, the reviews exhibit a slightly positive sentiment.

6.5.3 15 Most common words in summary and their polarity



Top 15 Common Words in summary and their Polarity (without stopwords)

Word: stars, Polarity: 0.0

Word: five, Polarity: 0.0

Word: game, Polarity: -0.4

Word: great, Polarity: 0.8

Word: fun, Polarity: 0.3

Word: love, Polarity: 0.5

Word: good, Polarity: 0.7

Word: four, Polarity: 0.0

Word: like, Polarity: 0.0

Word: expansion, Polarity: 0.0

Word: kids, Polarity: 0.0

Word: cute, Polarity: 0.5

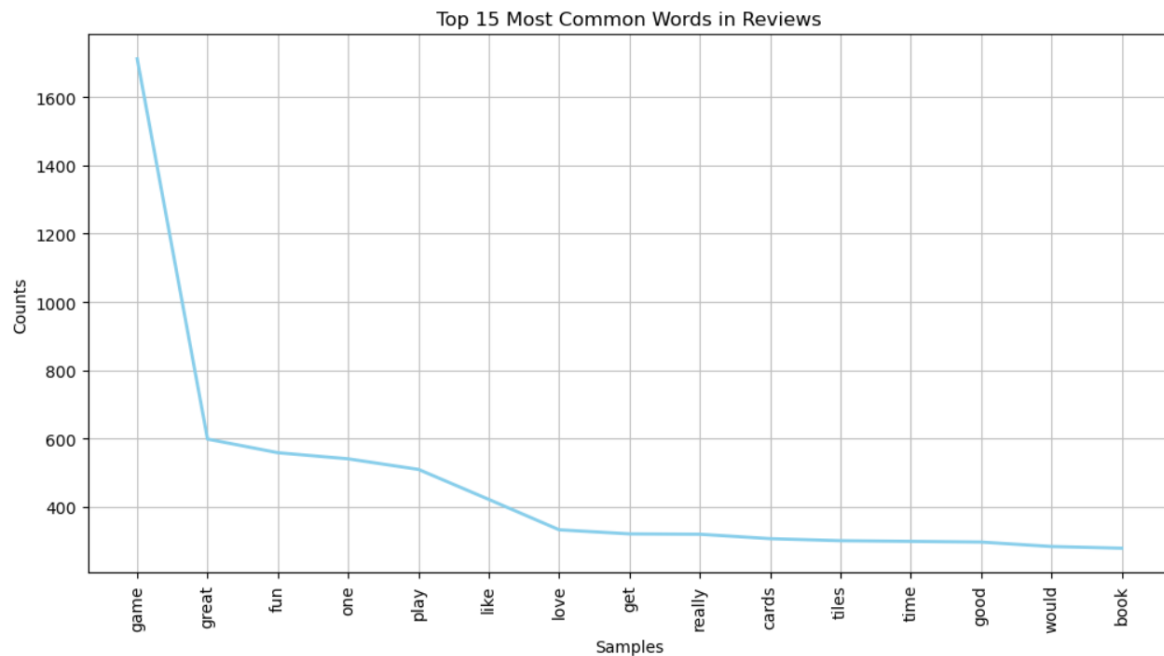
Word: book, Polarity: 0.0

Word: one, Polarity: 0.0

Word: old, Polarity: 0.1

Positive words such as "great," "good," "love," and "cute" suggest that the overall sentiment in the summaries is generally favourable. The word "game" with a negative polarity indicates there might be some critiques or negative feedback related to the game experience.

6.5.4 15 most common words in review and their polarity



Top 15 Common Words in review and their Polarity (without stopwords)

Word: game, Polarity: -0.4

Word: great, Polarity: 0.8

Word: fun, Polarity: 0.3

Word: one, Polarity: 0.0

Word: play, Polarity: 0.0

Word: like, Polarity: 0.0

Word: love, Polarity: 0.5

Word: get, Polarity: 0.0

Word: really, Polarity: 0.2

Word: cards, Polarity: 0.0

Word: tiles, Polarity: 0.0

Word: time, Polarity: 0.0

Word: good, Polarity: 0.7

Word: would, Polarity: 0.0

Word: book, Polarity: 0.0

Overall, the words with positive polarity dominate the list, with words like "great," "love," "fun," and "good" reflecting a favourable tone in the data. The word "game" having a negative polarity suggests there may be some critiques or less favourable mentions of games.

7. Answer the problem question

Problem question	Actionable Insights	Business Recommendation
How can Turtle games use the customer trends to improve overall sales performance?	• There was a positive correlation between remuneration and spending score with loyalty points, suggesting that higher earning customers and higher spending customers tend to accumulate more loyalty points. • Additionally, grouping the data by spending score revealed that customers with higher spending scores also had higher loyalty points, confirming that spending behaviour is a key factor in loyalty point accumulation.	• In order to improve overall sales performance, Turtle games should provide benefits to the customer for accumulating loyalty points. • For instance, customers could get different levels of discount (5%, 10%, 15%, 20%) for purchasing products in exchange for the loyalty points. • In this way, customers see benefit in accumulating loyalty points and since accumulating loyalty points is directly correlated with customer spending, Turtle games can improve overall sales performance by incentivising the customers to accumulate more loyalty points.